

Software Requirements Specification (SRS)

Project Title: TrackFinder

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Date:

Author:

1. Introduction

In universities, students are often faced with multiple Career tracks, specializations, and learning paths. Choosing the appropriate track can be challenging, especially when information about available options is scattered or not clearly organized. Many students struggle to identify the track that **best matches** their **interests, skills, and career goals**.

1.1 Purpose

The purpose of this document is to describe the requirements of the system called **TrackFinder**.

1.2 Scope

The system is designed to help students to find the right track that matches their interests, skills, and career goals.

It allows users to explore and discover different career tracks available, view detailed information about each track, and identify the track that best matches their interests and career goals.

The system will provide organized track information, including track descriptions, requirements, related sections, and guidance that helps students make informed decisions about their career paths.

2. Overall Description

TrackFinder is designed to assist university students in discovering and exploring different Career tracks in a clear and structured way. The system provides a centralized platform

where students can browse available tracks, understand their requirements, and view detailed information about each track.

Through an organized structure that group tracks into sections, it helps students make more informed decisions about their Career paths while also allowing administrators to efficiently manage and update track information.

The main goal of the system is to improve **accessibility to Career track information, reduce confusion among students**, and support better decision-making regarding their future specialization.

2.1 Product Perspective

TrackFinder is a web-based system designed to help university students explore and understand the different career tracks available.

The system provides a centralized platform where track information is organized and easily accessible.

TrackFinder is designed to simplify the process of discovering career tracks by presenting them in a structured hierarchy, such as sections and tracks, allowing students to navigate and explore available options efficiently.

The system will interact with a database that stores information about tracks, sections, and users.

2.2 Product Functions

The main functions of the TrackFinder system include:

- Allow students to browse and explore available career tracks.
- Provide detailed information about each track.
- Organize tracks into sections for easier navigation.
- Allow administrators to add, update, and manage tracks.
- Allow users to search for tracks based on their interests.
- Provide clear and structured information about career paths.

2.3 User Classes and Characteristics

Student:

University students who want to explore available career tracks and learn more about them before choosing their specialization.

Administrator:

Responsible for managing the system, including adding new tracks, updating track information, and organizing tracks into sections

Instructor:

Responsible for creating and managing educational content related to career tracks.

Instructors can add courses, learning materials, and resources that help students follow the learning roadmap for each track.

They ensure that the content is accurate, updated, and aligned with the required skills for each career track.

2.4 Operating Environment

The system will operate as a web application.

Technology: ASP.NET MVC app.

Database: SQL Server.

The system will be accessible through modern web browsers such as Chrome, Edge, and Firefox.

2.5 Design Constraints

- The system must be implemented as a web application.
- The system must use a relational database for storing data.
- The system must provide a simple and user-friendly interface.

3. System Features:-

3.1 Student Assessment

Students can take a short assessment to identify their interests and skills. The system analyzes the answers to determine which academic or career tracks suit them best.

3.2 Track Recommendation

Based on the assessment results, the system recommends the most compatible tracks, ranking them from most to least suitable.

3.3 Nature of the Job

For each recommended track, the system provides information about the job's nature, including daily tasks and key responsibilities.

3.4 Work Environment

The system shows the expected work environment for each track, such as office-based, field work, remote, or hybrid settings.

3.5 Salary Range

Provides an idea of the expected salary range for each track, helping students understand the financial aspects of the role.

3.6 Required Skills and Tools

Lists the key skills and tools that students need to learn to succeed in each track, including both technical and soft skills.

3.7 Learning Roadmap

Offers a structured learning path for each track, from basic skills to advanced ones, so students can follow clear steps to reach their goals.

3.8 Learning Resources

Provides reliable educational resources such as books, online courses, and practical tools to help students acquire the required skills.

3.9 Reporting

The system generates reports on assessments and results, helping administrators analyze student trends and track engagement.

4. External Interface Requirements

4.1 User Interfaces

TrackFinder will provide a **user-friendly web interface** accessible through modern browsers. The interface will be intuitive, visually clean, and easy to navigate for university students.

Main pages include:

- **Home Page:** Introduction to the system and navigation to other sections.
- **Login Page:** Secure login for students and administrators.

- **Registration Page:** Allows new students to create an account.
- **Dashboard:** Overview of student's recommended tracks and progress.
- **Assessment Page:** Interface for taking the assessment test.
- **Track Details Page:** Shows detailed information for each recommended track.
- **Profile Page:** Student information and progress summary.

4.2 Software Interfaces

The system interacts with:

- **Database Management System:** SQL Server to store students, tracks, and assessment results.
- **Web Browsers:** Chrome, Edge, Firefox, or any modern browser.
- **External APIs (if any):** For additional learning resources or skill databases.

4.4 Communication Interfaces

- Communication between client and server will use **HTTP/HTTPS protocols**.
- Data exchange will be in **JSON format** via **RESTful APIs**.

5. Non-Functional Requirements

5.1 Performance

- The system should support at least **1000 concurrent users** without noticeable performance degradation.
- Response time for loading pages or recommendations should not exceed **3 seconds**.

5.2 Security

- All passwords must be **hashed and stored securely**.
- User authentication and session management must be implemented to prevent unauthorized access.
- All sensitive data must be encrypted in transit using **HTTPS**.

5.3 Reliability

- The system should have **99% uptime**, ensuring it is available to students whenever needed.
- Backup mechanisms must be in place to prevent data loss.

5.4 Usability

- The system should be intuitive and easy for students to navigate without training.

- Interface elements should be consistent across pages.
- Assessment and track recommendation processes should be straightforward and visually guided.

6. Database Requirements

Main Entities

The database will store information about students, tracks, assessments, and administrators.

Main entities include:

- **Users** (Students & Administrators)
- **Tracks**
- **Sections**
- **Assessments**
- **Assessment Responses**

7. Future Enhancements

- **Advanced Analytics:** Provide deeper insights into student assessments and track trends.
- **Integration with External Learning Platforms:** Automatically link skills and tools to online courses or resources.
- **Gamification:** Add badges, progress tracking, and achievements to motivate students.